

Detailed Marking Report — Mathematics

exam-11-ch1-2-4-5-7 (Version 2) | Units 1, 2, 4, 5, 7 | Federal Board Annual Examination 2026
Candidate: Seemab | Attempt date: 3 July 2026 | Marked: 3 July 2026 (independently deep-check verified)

58.5/65

TOTAL SCORE

90.0%

PERCENTAGE

A+

GRADE

12/12

SECTION A (MCQ)

31.5/33

SECTION B

15/20

SECTION C

All three sections present and clearly readable across 13 handwritten pages. Seemab answered Section B first (pages 1-7), then Section C (pages 8-12), then listed all 12 MCQ answers on page 13, which is allowed. All 11 Section B questions and all 4 Section C questions were attempted, except Q3(iii)(b) which was left blank.

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

MCQ answers listed on page 13. Every letter matches the Version 2 key exactly — a clean 12/12.

#	Topic	Key	Seemab	Result	Mark
1	Quotient rule of exponents	B	B	✓	1
2	$\sqrt{a}$ for $a > 0$ is non-negative real	C	C	✓	1
3	Scientific notation of 0.000307	B	B	✓	1
4	$\log_3 81$	C	C	✓	1
5	$\log 18$ via $\log 2$ and $\log 3$	B	B	✓	1
6	Factorization of $x^2 - 9x + 20$	A	A	✓	1
7	$\text{HCF} \times \text{LCM} = P \times Q$	C	C	✓	1
8	Standard form of linear equation	B	B	✓	1
9	Solution set of $ 3x - 6 = 9$	A	A	✓	1
10	Distance (0,0) to (5,12)	C	C	✓	1
11	Midpoint of A(4,-2), B(-2,8)	B	B	✓	1
12	HCF of $x^2 - 4$ and $x^2 - x - 2$	B	B	✓	1

Section A Total

12 / 12

Perfect 12/12 on MCQs. This extends the current consecutive-perfect MCQ streak to 11 exams (attempts 14-24). Math MCQ all-time: 71/72.

Section B — Short Questions (11 × 3 = 33 Marks)

Seemab answered the OR variant for Q2(i), Q2(iii); the main for all others. Ten of eleven questions were correct or near-perfect.

Q2(i) OR — Equivalent radicals: (i) $(-8)^{1/3}$ (ii) $(\sqrt{25})^4$ (iii) $(x^{-3})^2 \cdot (x^0)^5$

3/3

(i) $(-8)^{1/3} = -2$. ✓ Correct (1/1)

(ii) $(\sqrt{25})^4 = 5^4 = 625$. ✓ Correct (1/1)

(iii) $(x^{-3})^2 \cdot (x^0)^5 = x^{-6} = 1/x^6$. ✓ Correct (1/1)

Q2(ii) main — True/False with counterexample

2.5/3

(a) Every integer is rational — TRUE. ✓ (1/1)

(b) Every irrational is real — TRUE. ✓ (1/1)

(c) Every rational is an integer — FALSE. -0.5: correctly marked False, but the required counterexample (e.g. $1/2$ is rational but not an integer) was not written. (0.5/1)

Q2(iii) OR — Notation conversion

3/3

(i) $7.05 \times 10^{-4} = 0.000705$. ✓ (1/1)

(ii) $2.31 \times 10^6 = 2,310,000$. ✓ (1/1)

(iii) $0.0000000512 = 5.12 \times 10^{-8}$. ✓ (1/1)

Q2(iv) main — Find x

2.5/3

(i) $\log_x 125 = 3 \Rightarrow x = 5$. ✓ (1/1)

(ii) $\log_4 x = -3/2 \Rightarrow x = 1/8$. ✓ (1/1)

(iii) $\log_2(x^2-3) = \log_2 13 \Rightarrow x^2 = 16 \Rightarrow x = 4$. *-0.5: only $x = 4$ given; the key answer is $x = \pm 4$ (both satisfy the domain). (0.5/1)*

Q2(v) main — Log using tables

3/3

(i) $\log 4580 = 3.6609$. ✓ (1/1)

(ii) $\log 0.00093 = \bar{4}.9685$. ✓ (1/1)

(iii) $\log 54 = 1.7324$. ✓ (1/1)

Q2(vi) main — Factorize completely

3/3

(i) $x^4+4m^4 = (x^2+2m^2-2mx)(x^2+2m^2+2mx)$. ✓ (1.5/1.5)

(ii) $3x^2+11x+6 = (3x+2)(x+3)$. ✓ (1.5/1.5)

Q2(vii) main — HCF by factorization

2.5/3

(i) $x^2-49 = (x-7)(x+7)$. ✓ (1/1)

(ii) $x^2-4x-21$: *-0.5: mis-copied the constant as +21 and factored to $(x-3)(x-7)$, which does not equal $x^2-4x-21$ [correct is $(x-7)(x+3)$]. (0.5/1)*

HCF = $(x-7)$. ✓ *The final HCF is correct despite the slip above. (1/1)*

Q2(viii) main — Solve $(2x+3)/5 = (3-4x)/8$

3/3

Cross-multiplied $8(2x+3)=5(3-4x) \Rightarrow 36x=-9 \Rightarrow x = -1/4$. ✓ *Correct (3/3)*

Q2(ix) main — Solve $|4x-3| = 13$

3/3

Both cases solved: $x = 4$ and $x = -5/2$. Solution set $\{4, -5/2\}$. ✓ *Correct (3/3)*

Q2(x) main — Collinearity of A(1,-2), B(7,8), C(4,3)

3/3

$AB = 2\sqrt{34}$, $BC = \sqrt{34}$, $AC = \sqrt{34}$; $BC + AC = AB \Rightarrow$ points collinear. ✓ *Correct (3/3)*

Q2(xi) main — Midpoint application

3/3

P(3,k) midpoint of A(-1,7), B(7,-3) $\Rightarrow k = 2$; then $PA = \sqrt{41}$ units. ✓ *Correct (3/3)*

Section B Total

31.5 / 33

Section C — Long Questions (4 × 5 = 20 Marks)

Seemab answered the OR variant for Q3(i) and Q3(ii); the main for Q3(iii) and Q3(iv). This is where the marks were lost.

Q3(i) OR — Evaluate $x = {}^3\sqrt{(8.59) \times (55.6)^2 \div (2.51 \times \sqrt{2.12})}$ using log tables

4.5/5

Expansion $\log x = (1/3)\log 8.59 + 2 \log 55.6 - \log 2.51 - (1/2)\log 2.12$ set up correctly. ✓ Table values (0.9339, 1.7450, 0.3996, 0.3263) all correct. ✓

Final **antilog ≈ 1732 is correct** (true value = 3.2386 $\rightarrow$ 1732). ✓

-0.5: one intermediate combining line reads 3.56545, which is arithmetically wrong (the two negative terms sum to 0.56275, not 0.23585). She recovered the correct antilog regardless, but the working is inconsistent.

Q3(ii) OR — (a) $\sqrt{(4a^2+12ab+9b^2)}$ (b)(i) $(x^2+3x-4)(x^2+3x+5)+8$ (b)(ii) $8a^3-125b^3-2a+5b$

4.5/5

(a) Recognised $4a^2+12ab+9b^2 = (2a+3b)^2$. *-0.5: stopped at the perfect-square form; did not write the square root itself, $\pm(2a+3b)$. (1/1.5)*

(b)(i) Substituting $a = x^2+3x$: $(a-4)(a+5)+8 = (a+4)(a-3) \rightarrow (x^2+3x+4)(x^2+3x-3)$. ✓ *Correct (2/2)*

(b)(ii) $(2a)^3-(5b)^3-(2a-5b) = (2a-5b)(4a^2+10ab+25b^2-1)$. ✓ *Correct (1.5/1.5)*

Q3(iii) main — (a) Solve $(y+1)/3 + (y+1)/2 = 2 - (y+3)/2$ (b) Solve $(3/4)x - 1 \geq x + 1$ in set notation

1/5

(a) LHS combined correctly to $(5y+5)/6$. *-2: a sign error on the right side — $2 - (y+3)/2$ was written as $(4-y+3)/2 = (7-y)/2$ instead of $(4-y-3)/2 = (1-y)/2$. This led to $y = 2$; the correct answer is $y = -1/4$. Method credit only. (1/3)*

(b) *-2: not attempted. The inequality $(3/4)x-1 \geq x+1 \rightarrow x \leq -8$, set $\{x \mid x \in \mathbb{R} \wedge x \leq -8\}$, is absent from all pages. (0/2)*

Q3(iv) main — A(0,0), B(6,0), C(6,4), D(0,4): (a) sides not all equal (b) diagonals + midpoints $\rightarrow$ rectangle

5/5

(a) $AB=6$, $BC=4$, $CD=6$, $DA=4$; opposite sides equal, not all equal. ✓ (2/2)

(b) $AC = BD = \sqrt{52}$ (diagonals equal); midpoint of AC = midpoint of BD = (3,2) (bisect each other) $\rightarrow$ ABCD is a rectangle. ✓ *Complete and correct (3/3)*

Section C Total

15 / 20

Independent deep-check: a second, separate verifier re-marked every item from the raw scans and reconciled against this report. It confirmed all 12 MCQ letters, that Q3(iii)(b) is genuinely blank, that the antilog 1732 in Q3(i) is correct, that the Q2(vii) and Q3(iii)(a) errors are real, and that the arithmetic reconciles to 58.5/65. No changes were made on review.

Final Score Breakdown

Section	Available	Earned	%	Notes
Section A — MCQ	12	12	100%	All 12 correct — 11th consecutive perfect MCQ
Section B — Short Questions	33	31.5	95.5%	Lost 1.5: small slips on Q2(ii), Q2(iv), Q2(vii)
Section C — Long Questions	20	15	75.0%	Lost 5: mostly Q3(iii) — sign error in (a), (b) unattempted
TOTAL	65	58.5	90.0%	Grade A+

Deduction Log (all 6.5 marks lost)

Question	Sub-part	Issue	Lost
Q3(iii)	(a) equation	Sign error: $2-(y+3)/2$ taken as $(7-y)/2$ instead of $(1-y)/2$, giving $y=2$ instead of $y=-1/4$. Method credited.	-2
Q3(iii)	(b) inequality	Not attempted. $(3/4)x-1 \geq x+1 \rightarrow x \leq -8$ with set notation never written.	-2
Q2(ii)	(c) counterexample	Correctly marked False but omitted the required counterexample ($1/2$ rational, not integer).	-0.5
Q2(iv)	(iii) $x=\pm 4$	Gave only $x=4$; missed the negative root $x=-4$.	-0.5
Q2(vii)	2nd factorization	Mis-copied $x^2-4x-21$ as $+21$ and factored to $(x-3)(x-7)$; HCF still correct.	-0.5
Q3(i)	arithmetic line	One combining line (3.56545) is wrong, though the final antilog 1732 is correct.	-0.5
Q3(ii)	(a) square root	Left answer as $(2a+3b)^2$ without writing the root $\pm(2a+3b)$.	-0.5

Strengths in this exam

- Perfect MCQ (12/12) — all five topic areas covered cleanly, 11th straight perfect score
- Distance, midpoint and collinearity work in coordinate geometry is flawless (Q2(x), Q2(xi), Q3(iv) all perfect)
- Rectangle proof Q3(iv) fully complete: sides, equal diagonals, and bisecting midpoints all shown — exactly the rigour that was missing on Math exam-10
- Log-table evaluation Q3(i) landed the correct antilog 1732 with a clean expansion and correct table lookups
- Factorization is strong: x^4+4m^4 by add-and-subtract, sum/difference of cubes with the trailing linear term, and the substitution trick on $(x^2+3x-4)(x^2+3x+5)+8$ all correct
- Absolute-value and linear-equation solving (Q2(viii), Q2(ix)) clean and complete with solution sets
- Scientific-notation and log-table basics in Section B all correct

Errors in this exam

- **Q3(iii)(a):** a single sign slip on the right-hand side flipped the whole answer. When distributing a minus over $(y+3)/2$, the -3 became $+3$. Costliest conceptual error on the paper; the algebra either side of it was fine.
- **Q3(iii)(b):** left blank. This is the recurring "stopped before all sub-parts" pattern — a 2-mark inequality that was well within reach. Same theme flagged on Math exam-10 and the last two Physics papers.
- **Q3(ii)(a):** recognised the perfect square but did not take the final square root. One extra line $(\pm(2a+3b))$ would have secured it.
- **Small Section B slips:** missing counterexample (Q2(ii)), missing negative root $x=-4$ (Q2(iv)), and a mis-copied constant in Q2(vii). Each is half a mark; together they are the difference between 90% and 92%.